Test Plan Document

Hospital Management System

Test case naming convention: UT = Unit Testing, IT = Integration Testing, ST = System Testing (e.g., UT_M1_01 = Unit Test, Module 1, Test Case #1).

Module: User Authentication & Role Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M1_01	User Authentication & Role Management	Verify successful login with valid Admin credentials	Admin account exists in the system	1: Navigate to login page 2: Enter valid Admin username and password 3: Click Login	Username: admin01 Password: Admin@123	Admin dashboard is displayed successfully		
UT_M1_02	User Authentication & Role Management	Verify login fails with invalid password	Admin account exists in the system	1: Navigate to login page 2: Enter valid username with incorrect password 3: Click Login	Username: admin01 Password: wrongpass	Error message 'Invalid credentials' is displayed		
UT_M1_03	User Authentication & Role Management	Verify login fails with blank username/password	Login page is accessible	1: Navigate to login page 2: Leave username and password fields blank 3: Click Login	Username: (blank) Password: (blank)	Validation message 'Username and Password are required' is displayed		
UT_M1_04	User Authentication & Role Management	Verify Doctor login redirects to Doctor dashboard	Doctor account exists in the system	1: Navigate to login page 2: Enter valid Doctor credentials 3: Click Login	Username: drsmith Password: Doctor@123	Doctor-specific dashboard with appointments and patient list is displayed		
UT_M1_05	User Authentication & Role Management	Verify Receptionist login redirects to Receptionist dashboard	Receptionist account exists	1: Navigate to login page 2: Enter valid Receptionist credentials 3: Click Login	Username: reception01 Password: Recep@123	Receptionist dashboard with appointment booking options is displayed		
UT_M1_06	User Authentication & Role Management	Verify Patient login redirects to Patient dashboard	Patient account exists	1: Navigate to login page 2: Enter valid Patient credentials 3: Click Login	Username: patient01 Password: Pat@1234	Patient dashboard with appointment history and medical records is displayed		
IT_M1_01	User Authentication & Role Management	Verify role-based access control restricts unauthorized module access	Receptionist logged in	1: Login as Receptionist 2: Attempt to access Billing/Reports admin-only module via URL	Role: Receptionist	Access denied message is displayed and user is redirected to dashboard		
IT_M1_02	User Authentication & Role Management	Verify session expires after logout	User is logged in	1: Login with valid credentials 2: Click Logout 3: Click browser back button	Any valid user credentials	User is redirected to login page and cannot access previous session		
ST_M1_01	User Authentication & Role Management	Verify account lockout after multiple failed login attempts	Account exists and lockout policy is configured	1: Enter incorrect password 5 times consecutively 2: Attempt a 6th login	Username: admin01 Password: wrongpass (x5)	Account is locked and message 'Account locked, try after some time' is displayed		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_M1_02	User Authentication & Role Management	Verify system displays role-specific dashboard immediately after login for all roles	Valid accounts exist for Admin, Doctor, Receptionist, Patient	1: Login sequentially with each role's credentials 2: Observe dashboard loaded for each	Credentials for all four roles	Each role sees only its corresponding dashboard and menu options		

Module: Patient & Medical Record Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M2_01	Patient & Medical Record Management	Verify new patient registration with valid details	User has Receptionist/Admin access	1: Navigate to Patient Registration 2: Enter patient name, age, gender, contact, address 3: Click Save	Name: Ravi Kumar Age: 34 Gender: Male Contact: 9876543210	Patient is registered successfully and a unique Patient ID is generated		
UT_M2_02	Patient & Medical Record Management	Verify registration fails with missing mandatory fields	Patient registration form is open	1: Leave 'Name' field blank 2: Fill other fields 3: Click Save	Name: (blank) Age: 28	Validation error 'Name is required' is displayed		
UT_M2_03	Patient & Medical Record Management	Verify registration fails with invalid contact number format	Patient registration form is open	1: Enter contact number with letters 2: Click Save	Contact: 98AB56CD10	Validation error 'Enter a valid contact number' is displayed		
UT_M2_04	Patient & Medical Record Management	Verify patient medical history can be added	Patient is already registered	1: Open patient profile 2: Navigate to Medical History tab 3: Add diagnosis and treatment details 4: Click Save	Diagnosis: Hypertension Treatment: Medication	Medical history entry is saved and displayed in patient profile		
UT_M2_05	Patient & Medical Record Management	Verify prescription can be added to a patient record	Doctor is logged in and patient record exists	1: Open patient profile 2: Navigate to Prescription tab 3: Enter medicine name, dosage, duration 4: Click Save	Medicine: Paracetamol Dosage: 500mg twice daily	Prescription is saved and linked to patient record		
IT_M2_01	Patient & Medical Record Management	Verify searching a patient by Patient ID retrieves correct record	Patient with known ID exists	1: Navigate to Patient Search 2: Enter Patient ID 3: Click Search	Patient ID: PT1023	Correct patient record with details is displayed		
IT_M2_02	Patient & Medical Record Management	Verify updated patient details reflect across doctor and receptionist views	Patient record exists	1: Update patient contact number as Receptionist 2: Login as Doctor and open same patient profile	New Contact: 9123456780	Updated contact number is visible in Doctor's view of the patient		
IT_M2_03	Patient & Medical Record Management	Verify duplicate patient registration with same contact number is flagged	A patient with the same contact number already exists	1: Register a new patient using an existing contact number 2: Click Save	Contact: 9876543210 (existing)	System warns of a possible duplicate patient record		
ST_M2_01	Patient & Medical Record Management	Verify complete patient record (details + history + prescriptions) displays correctly end-to-end	Patient has registration, history and prescription data	1: Login as Doctor 2: Open patient profile 3: Review all tabs	Existing Patient ID: PT1023	All patient information is displayed accurately and consistently		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_M2_02	Patient & Medical Record Management	Verify medical records remain accessible after multiple concurrent updates	Two staff members access same patient record	1: Receptionist updates contact info 2: Doctor simultaneously adds diagnosis 3: Refresh patient record	Existing Patient ID: PT1023	Both updates are saved without data loss or conflict		

Module: Doctor & Department Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M3_01	Doctor & Department Management	Verify new doctor can be added with valid details	Admin is logged in	1: Navigate to Doctor Management 2: Enter doctor name, specialization, contact 3: Click Save	Name: Dr. Anita Rao Specialization: Cardiology	Doctor is added successfully and listed in doctor directory		
UT_M3_02	Doctor & Department Management	Verify department creation with valid name	Admin is logged in	1: Navigate to Department Management 2: Enter department name 3: Click Save	Department: Cardiology	New department is created and listed		
UT_M3_03	Doctor & Department Management	Verify doctor addition fails without specialization field	Doctor Management form is open	1: Enter doctor name only 2: Leave specialization blank 3: Click Save	Name: Dr. John Specialization: (blank)	Validation error 'Specialization is required' is displayed		
UT_M3_04	Doctor & Department Management	Verify doctor availability schedule can be set	Doctor profile exists	1: Open doctor profile 2: Set available days and time slots 3: Click Save	Available: Mon-Fri, 9AM-1PM	Availability schedule is saved and reflected in the profile		
IT_M3_01	Doctor & Department Management	Verify assigning a doctor to a department updates department roster	Doctor and department both exist	1: Open doctor profile 2: Assign to 'Cardiology' department 3: Click Save 4: Open department view	Doctor: Dr. Anita Rao Department: Cardiology	Doctor appears under Cardiology department's doctor list		
IT_M3_02	Doctor & Department Management	Verify doctor list filters correctly by department	Multiple doctors assigned to different departments	1: Navigate to Doctor Directory 2: Apply filter for 'Orthopedics'	Filter: Orthopedics	Only doctors belonging to Orthopedics department are displayed		
IT_M3_03	Doctor & Department Management	Verify doctor availability reflects correctly on appointment booking screen	Doctor's availability is configured	1: Navigate to Appointment Booking 2: Select the doctor	Doctor: Dr. Anita Rao	Only the configured available time slots are shown as bookable		
ST_M3_01	Doctor & Department Management	Verify deactivating a doctor removes them from active booking list but retains historical records	Doctor has past appointments	1: Deactivate doctor profile 2: Attempt new appointment booking 3: Check historical patient records	Doctor: Dr. Anita Rao (deactivated)	Doctor no longer appears for new bookings; past records remain intact		
ST_M3_02	Doctor & Department Management	Verify department deletion is blocked if doctors are still assigned	Department has active doctors assigned	1: Attempt to delete a department with assigned doctors	Department: Cardiology (has doctors)	System prevents deletion and displays an appropriate warning message		

Module: Appointment Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M4_01	Appointment Management	Verify appointment can be booked for an available doctor slot	Patient and doctor exist; slot is available	1: Navigate to Book Appointment 2: Select doctor, date and available time slot 3: Click Confirm	Doctor: Dr. Rao Date: next available Slot: 10:00 AM	Appointment is booked successfully and confirmation is shown		
UT_M4_02	Appointment Management	Verify booking fails for an already occupied slot	Slot already booked by another patient	1: Select doctor and an already booked slot 2: Click Confirm	Slot: 10:00 AM (occupied)	Error message 'Slot not available' is displayed		
UT_M4_03	Appointment Management	Verify appointment can be rescheduled to a new available slot	Existing appointment is booked	1: Open existing appointment 2: Select 'Reschedule' 3: Choose a new available slot 4: Click Confirm	Original Slot: 10:00 AM New Slot: 2:00 PM	Appointment is updated to the new slot and confirmation is displayed		
UT_M4_04	Appointment Management	Verify appointment can be cancelled successfully	Existing appointment is booked	1: Open existing appointment 2: Click Cancel 3: Confirm cancellation	Appointment ID: APT2045	Appointment status changes to 'Cancelled' and slot becomes available		
UT_M4_05	Appointment Management	Verify booking fails when past date is selected	Booking screen is open	1: Select a date earlier than current date 2: Click Confirm	Date: past date	Validation error 'Cannot book appointment for a past date' is displayed		
IT_M4_01	Appointment Management	Verify cancelled appointment slot becomes available for other patients	An appointment was cancelled	1: Cancel an existing appointment 2: Login as a different patient 3: Attempt to book the freed slot	Freed Slot: 10:00 AM	New patient is able to successfully book the freed slot		
IT_M4_02	Appointment Management	Verify appointment booking updates doctor's daily schedule view	Doctor account exists	1: Patient books an appointment with the doctor 2: Login as the doctor 3: Open daily schedule	Doctor: Dr. Rao	New appointment appears correctly in doctor's schedule		
ST_M4_01	Appointment Management	Verify end-to-end appointment workflow: book, reschedule, notify, and complete	Patient and doctor accounts exist	1: Book an appointment 2: Reschedule it once 3: Verify reminder notification is sent 4: Mark appointment as completed after consultation	Doctor: Dr. Rao Patient: Ravi Kumar	Appointment progresses correctly through all statuses with accurate records		
ST_M4_02	Appointment Management	Verify concurrent booking attempts for the same slot by two patients are handled without double-booking	One slot available; two patients attempt simultaneously	1: Patient A and Patient B both select the same slot 2: Both click Confirm at nearly the same time	Slot: 11:00 AM	Only one booking succeeds; the other receives a 'Slot no longer available' message		

Module: Admission & Discharge Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M5_01	Admission & Discharge Management	Verify patient admission can be recorded with room/bed allocation	Patient is registered; bed is available	1: Navigate to Admission 2: Select patient, ward, and bed 3: Click Admit	Patient: Ravi Kumar Ward: General Bed: G-12	Patient is marked as admitted and bed is allocated		
UT_M5_02	Admission & Discharge Management	Verify admission fails when selected bed is already occupied	Bed is already occupied	1: Select an occupied bed for admission 2: Click Admit	Bed: G-12 (occupied)	Error message 'Bed already occupied' is displayed		
UT_M5_03	Admission & Discharge Management	Verify treatment details can be added during hospitalization	Patient is currently admitted	1: Open admitted patient's record 2: Add treatment/procedure details 3: Click Save	Treatment: IV therapy	Treatment entry is saved under the patient's admission record		
UT_M5_04	Admission & Discharge Management	Verify patient discharge can be processed with discharge summary	Patient is currently admitted	1: Open admitted patient's record 2: Click Discharge 3: Enter discharge summary and date 4: Click Confirm	Discharge Date: current date Summary: Recovered, advised rest	Patient status changes to 'Discharged' and bed becomes available		
IT_M5_01	Admission & Discharge Management	Verify bed becomes available in real time after patient discharge	Patient is admitted to a bed	1: Discharge the patient 2: Navigate to bed availability view	Bed: G-12	Bed status updates to 'Available' immediately after discharge		
IT_M5_02	Admission & Discharge Management	Verify admission record links correctly to billing for room charges	Patient admitted for more than one day	1: Admit patient 2: Discharge after stay 3: Navigate to Billing module	Ward: General, Duration: 3 days	Room charges for the stay duration appear correctly in the bill		
ST_M5_01	Admission & Discharge Management	Verify complete admission-to-discharge workflow updates all related modules	Patient is registered and a bed is available	1: Admit patient 2: Add treatments 3: Discharge patient with summary 4: Verify bed status, medical record and billing	Patient: Ravi Kumar	All modules (bed management, medical record, billing) reflect consistent, accurate data		
ST_M5_02	Admission & Discharge Management	Verify system prevents discharge without completing mandatory discharge summary	Patient is currently admitted	1: Click Discharge 2: Leave discharge summary blank 3: Click Confirm	Discharge Summary: (blank)	Validation error 'Discharge summary is required' is displayed and discharge is blocked		

Module: Billing & Payment Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M6_01	Billing & Payment Management	Verify bill is generated correctly for a consultation	Patient has completed a consultation	1: Navigate to Billing 2: Select patient and consultation service 3: Click Generate Bill	Service: Consultation Fee: Rs.500	Bill is generated with correct amount and itemized details		
UT_M6_02	Billing & Payment Management	Verify bill correctly aggregates multiple services (tests, medicines, treatment)	Patient has multiple billable services	1: Select patient 2: Add consultation, lab test, and medicine charges 3: Click Generate Bill	Consultation: Rs.500 Lab Test: Rs.800 Medicine: Rs.300	Total bill amount equals the sum of all itemized charges (Rs.1600)		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M6_03	Billing & Payment Management	Verify payment can be recorded against a generated bill	Bill has been generated and is unpaid	1: Open the bill 2: Select payment mode and enter amount 3: Click Record Payment	Amount: Rs.1600 Mode: Cash	Payment is recorded and bill status updates to 'Paid'		
UT_M6_04	Billing & Payment Management	Verify partial payment updates the bill's balance due correctly	Bill total is Rs.1600 and unpaid	1: Open the bill 2: Enter a partial payment amount 3: Click Record Payment	Amount Paid: Rs.1000	Bill status shows 'Partially Paid' with balance due of Rs.600		
UT_M6_05	Billing & Payment Management	Verify payment entry fails with negative or zero amount	Bill is open for payment	1: Enter a negative or zero payment amount 2: Click Record Payment	Amount: -100	Validation error 'Enter a valid payment amount' is displayed		
IT_M6_01	Billing & Payment Management	Verify generated bill is linked correctly to the patient's medical record	Bill is generated for a patient	1: Generate a bill 2: Open patient's medical record 3: Navigate to Billing History tab	Patient: Ravi Kumar	The bill appears correctly under the patient's billing history		
IT_M6_02	Billing & Payment Management	Verify admission room charges are automatically included in the final bill	Patient was admitted and discharged	1: Discharge an admitted patient 2: Generate final bill	Room charges for 3 days	Room charges are automatically itemized in the generated bill		
ST_M6_01	Billing & Payment Management	Verify end-to-end billing workflow from service completion to payment receipt	Patient has completed consultation, tests and treatment	1: Generate consolidated bill 2: Record full payment 3: Generate/print payment receipt	Total Bill: Rs.2500	Receipt is generated accurately reflecting all services and payment		
ST_M6_02	Billing & Payment Management	Verify billing reports reconcile correctly with individual recorded payments over a period	Multiple bills and payments exist for the month	1: Generate monthly billing report 2: Cross-check total against individual bill/payment records	Period: current month	Report totals match the sum of individual payment records with no discrepancy		

Module: Notification & Reminder Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M7_01	Notification & Reminder Management	Verify appointment reminder notification is generated a day before the scheduled appointment	Patient has an appointment scheduled for the next day	1: Wait for/trigger the scheduled reminder job 2: Check patient's notification list	Appointment Date: tomorrow	Reminder notification for the appointment is displayed to the patient		
UT_M7_02	Notification & Reminder Management	Verify notification is sent when a lab test is scheduled	A lab test has been scheduled for a patient	1: Schedule a lab test for the patient 2: Check notification list	Test: Blood Test Date: upcoming	Notification confirming the scheduled test is generated		
UT_M7_03	Notification & Reminder Management	Verify payment due reminder is generated for an unpaid bill nearing due date	Unpaid bill exists with a due date approaching	1: Trigger reminder job for due payments 2: Check patient's notification list	Bill: Rs.1600 unpaid, due in 1 day	Payment due reminder notification is displayed		
UT_M7_04	Notification & Reminder Management	Verify no duplicate reminders are generated for the same event	A reminder has already been sent for an appointment	1: Re-trigger the reminder job for the same appointment	Appointment ID: APT2045	No duplicate reminder notification is created		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_M7_01	Notification & Reminder Management	Verify rescheduled appointment updates the associated reminder	Appointment reminder already exists	1: Reschedule the appointment to a new date/time 2: Check the reminder notification	New Date/Time: updated slot	Reminder reflects the updated appointment date and time		
IT_M7_02	Notification & Reminder Management	Verify cancelled appointment removes its pending reminder	Reminder is scheduled for a pending appointment	1: Cancel the appointment 2: Check notification list	Appointment ID: APT2045 (cancelled)	The associated reminder is removed/cancelled		
ST_M7_01	Notification & Reminder Management	Verify all notification types (appointment, test, payment) appear correctly together in the notification center	Patient has upcoming appointment, scheduled test and pending payment	1: Trigger all reminder jobs 2: Open patient's Notification Center	Multiple pending events	All relevant notifications are listed accurately and in chronological order		
ST_M7_02	Notification & Reminder Management	Verify notifications are correctly routed to the intended recipient role only	Multiple roles have pending notifications	1: Trigger reminders for a patient event and a staff event 2: Login as Patient and check notifications 3: Login as Receptionist and check notifications	Mixed events for Patient and Receptionist	Each role sees only the notifications relevant to them		

Module: Report Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_M8_01	Report Management	Verify patient report can be generated for a given date range	Patient records exist within the date range	1: Navigate to Reports > Patient Report 2: Select date range 3: Click Generate	Date Range: last 30 days	Report listing patients registered/treated in the range is generated		
UT_M8_02	Report Management	Verify doctor performance report shows correct appointment counts	Doctor has completed appointments in the period	1: Navigate to Reports > Doctor Report 2: Select doctor and period 3: Click Generate	Doctor: Dr. Rao Period: current month	Report shows accurate count of appointments handled by the doctor		
UT_M8_03	Report Management	Verify appointment report correctly filters by status (booked/completed/cancelled)	Appointments with mixed statuses exist	1: Navigate to Reports > Appointment Report 2: Filter by status 'Cancelled' 3: Click Generate	Status Filter: Cancelled	Only cancelled appointments are listed in the report		
UT_M8_04	Report Management	Verify laboratory test report lists all tests conducted in a period	Lab tests were conducted within the period	1: Navigate to Reports > Lab Test Report 2: Select period 3: Click Generate	Period: current month	Report accurately lists all lab tests conducted in the selected period		
UT_M8_05	Report Management	Verify payment report total matches sum of individual recorded payments	Multiple payments recorded in period	1: Navigate to Reports > Payment Report 2: Select period 3: Click Generate	Period: current month	Report total equals the sum of all individual payments recorded		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_M8_01	Report Management	Verify report generation reflects real-time updates from other modules	A new appointment was just booked	1: Book a new appointment 2: Immediately generate Appointment Report for today	Today's date	Newly booked appointment appears in the generated report		
IT_M8_02	Report Management	Verify report can be exported/downloaded in a readable format	A report has been generated	1: Generate any report 2: Click Export/Download	Report: Patient Report	Report is downloaded successfully in the expected file format		
ST_M8_01	Report Management	Verify admission report accurately reflects data across admission and discharge modules	Multiple patients admitted and discharged in the period	1: Navigate to Reports > Admission Report 2: Select period 3: Click Generate 4: Cross-check with admission/discharge records	Period: current month	Admission report matches actual admission and discharge records		
ST_M8_02	Report Management	Verify system handles report generation for a large date range without errors	Historical data exists for over a year	1: Select a date range spanning 12 months 2: Click Generate for Patient Report	Date Range: 12 months	Report generates successfully within acceptable time without errors		